



Given observed behavior, infer the most likely emotional state of another agent, using your own world model as a reference point, corrected by the divergence between your model and theirs.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \Pr(\theta | o, P, Q)$$

θ = latent emotional state

o = observed behavior/event

P = your world model

Q = inferred world model of the other agent

$$\Pr(\theta | o, P, Q) = \frac{\Pr(o | \theta, P, Q) \Pr(\theta | P, Q)}{\Pr(o | P, Q)}$$

$$1. o \perp P | \theta, Q$$

The observation is independent of the observer's

World model, given the emotional state and the other agent's world model

$\Rightarrow$ Once you know θ and Q , knowing P tells you nothing additional about what behaviour you observe

$$Q \rightarrow \hat{Q}$$

$$2. \Pr(\theta | P, Q) \approx \Pr(\theta | P, \hat{Q}), \Pr(o | \theta, Q) \approx \Pr(o | \theta, \hat{Q})$$

$$\therefore \Pr(\theta | o, P, Q) \propto \underbrace{\Pr(o | \theta, \hat{Q})}_{\text{what I see}} \cdot \underbrace{\Pr(\theta | P, \hat{Q})}_{\text{what I expect}}$$

Stage 2: Learning $\hat{Q}$ - where the KL Divergence Lives

$\hat{Q}$ is our estimate of the other agent's world model

1. Explain Observed Behavior
2. Stay close to own World model P

$$\mathcal{L}(\hat{Q}) = \mathbb{E}[\text{MSE}(o_t, \hat{o}_t)] + \lambda D_{KL}(P \parallel \hat{Q})$$

$\hat{o}_t = \mathbb{E}[o_t \mid \hat{Q}]$, predicted observation under $\hat{Q}$

$D_{KL}(P \parallel \hat{Q})$ penalizes from drifting too far from P

λ controls the tradeoff between fit and regularization

$$\nabla_{\hat{Q}} \mathcal{L} = \nabla_{\hat{Q}} \mathbb{E}[\text{MSE}(o_t, \hat{o}_t)] + \lambda \nabla_{\hat{Q}} D_{KL}(P \parallel \hat{Q})$$

Gradient Descent Update: $\hat{Q}_{b+1} = \hat{Q}_b - \alpha \nabla_{\hat{Q}} \mathcal{L}(\hat{Q}_b)$

$$\Rightarrow \hat{Q}_{b+1} = \hat{Q}_b - \alpha (\nabla_{\hat{Q}} \mathbb{E}[\text{MSE}] + \lambda \nabla_{\hat{Q}} D_{KL}(P \parallel \hat{Q}))$$

$$\nabla_{\hat{Q}} D_{KL}(P \parallel \hat{Q}) = -\mathbb{E}_{x \sim P} \left[\frac{\nabla_{\hat{Q}} \hat{Q}(x)}{\hat{Q}(x)} \right]$$